

Frames and matrix designs

Vanishing injectivity probability at $4d$ minus 5 complex phase measurements

Difficulty: challenging **Importance:** interesting to the community**Status:** Solution claimed

Rating rationale: The asymptotic typical-injectivity question needs quantitative algebraic and probabilistic control beyond positive failure probability; it informs phase-retrieval measurement design.

Original statement

For every integer $d \geq 2$, draw $A_d \in \mathbb{C}^{(4d-5) \times d}$ with independent standard complex Gaussian entries: the real and imaginary parts are independent $N(0, 1/2)$. Let p_d be the probability that

$$|A_d x| = |A_d y| \implies y = e^{i\theta} x \text{ for some } \theta \in \mathbb{R}$$

holds for all $x, y \in \mathbb{C}^d$, where absolute value is componentwise. Equivalently, p_d is the probability that the induced phase-retrieval map on vectors modulo global phase is injective.

Is

$$\lim_{d \rightarrow \infty} p_d = 0?$$

This is part (b) of Vinzant's conjecture as restated in Randomstrasse 101. It distinguishes injective exceptional measurement systems from typical matrices at a row count just below $4d-4$. Injectivity is the basic identifiability requirement before conditioning and stable numerical inversion can be addressed.

Solution claim — 14 September 2026

Zhangsong Li's [13-page manuscript](#), Theorem 1.4, claims the stronger bound

$$p_d \leq C/d \quad (d \geq 2)$$

for an absolute constant $C > 0$. This implies the displayed limit. The argument plants a two-dimensional ambiguity, compares its law in L^2 to a covariance-matched Gaussian law, and uses a quantitative implicit-function argument to turn the planted approximate ambiguity into an exact one.

This remains a **solution claim**, not a repository-verified resolution: the manuscript is the primary source, and independent review and the repository's isolated Linux Lean verification remain outstanding. The original target, permanent identifier, and canonical path are unchanged. The implementation route is recorded in the [formalisation plan](#).

Local Lean development — 27 September 2026

The [Lean solution](#) now proves the bound $p_d \leq C/d$ for every $d \geq 2$ and the original limit $p_d \rightarrow 0$, using the original Gaussian law and all-signals injectivity predicate. The local build and transitive axiom audit pass, using only standard Lean axioms and no proof placeholders in the solution. See the [final assembly](#), [source guide](#), and [local verification record](#). This development check does not replace independent statement review or the isolated Linux Comparator/kernel workflow; the catalog status is unchanged.

References

1. A. S. Bandeira et al., *Randomstrasse 101: Open Problems of 2025*, arXiv:2603.29571 (2026), Conjecture 19(b). [Paper](#).
2. C. Vinzant, *A small frame and a certificate of its injectivity*, SAMPTA 2015, arXiv:1502.04656. The explicit 11-vector frame in $\mathbb{C}^4$ demonstrates why the below- $4d-4$ regime cannot simply be dismissed. [Paper](#).
3. Z. Li, *Resolution of Vinzant's Conjecture on Phase Retrieval Injectivity* (13 September 2026), Theorem 1.4. The manuscript claims $p_d \leq C/d$ for every $d \geq 2$. [Manuscript](#).
4. Z. Li, *On Injectivity of Phase Retrieval*, arXiv:2606.17922 (2026). This earlier version established only $p_d < 1$. [Paper](#).

Historical status check — 2026-09-10

Searched “Vinzant conjecture injectivity probability limit”, “phase retrieval 4M-5 2026”, and checked the latest June 2026 paper abstract/version. Its positive probability of noninjectivity does not supply an asymptotic lower bound tending to one. The already proved assertion $p_d < 1$ is deliberately excluded from this problem.

Audit update (2026-09-10): Rechecked Li’s June 2026 abstract and searched for the asymptotic part of Vinzant’s conjecture. The reported strict inequality $p_d < 1$ does not force $p_d \rightarrow 0$; this entry contained only the latter target. This was a bounded literature check, not a proof that no solution existed at that time.